V-Track Pro

Project Model

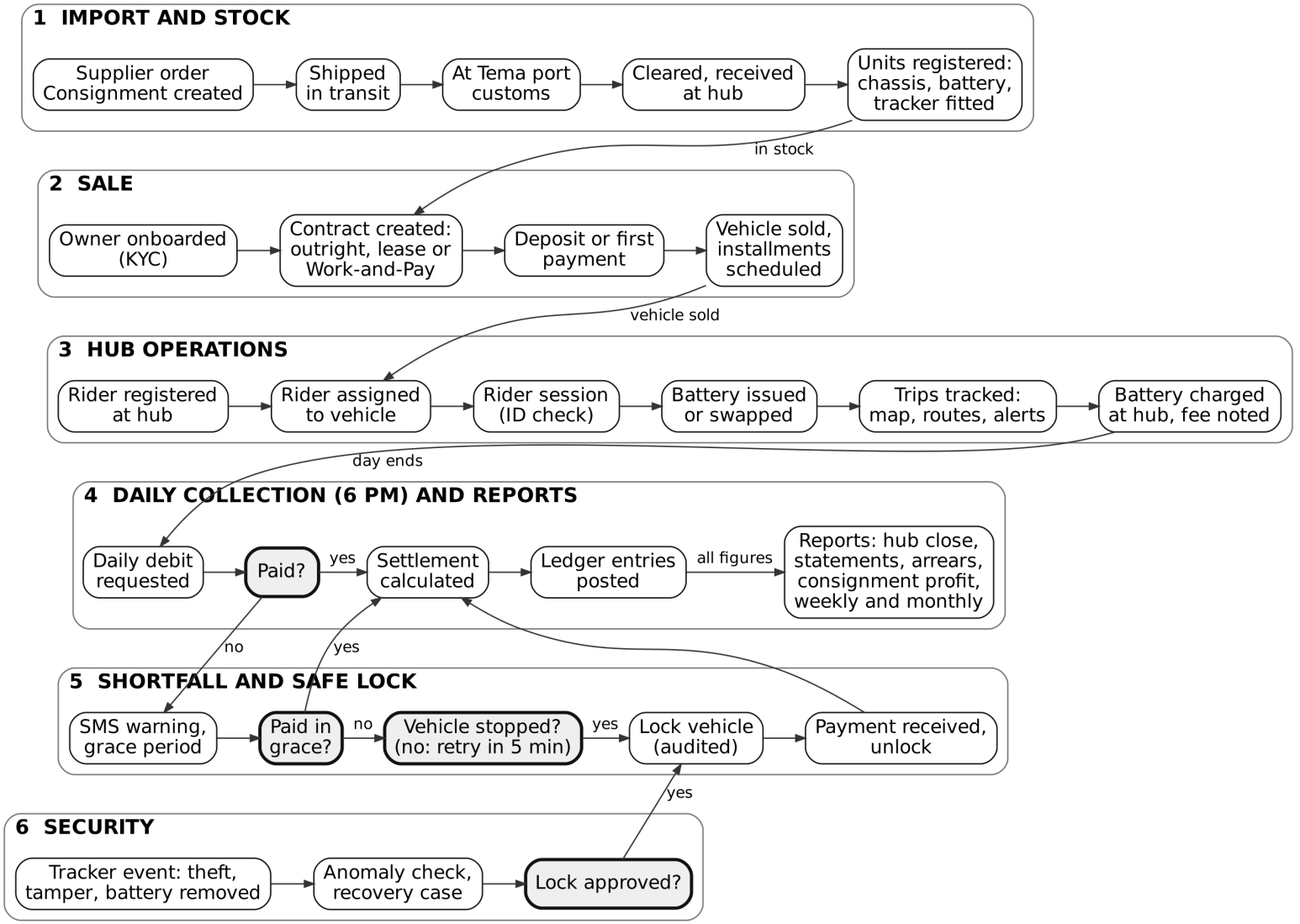
Domain, Lifecycles, Money and Rules

Electric bikes and tricycles, Ghana

Version 1.0 | 7 November 2026

Design document. Items marked as assumptions need the client's confirmation.

The Whole System on One Page



Read each row left to right. Rows connect downward: stock becomes a sale, a sale becomes an assigned rider, a day of work becomes a collection, and everything lands in the ledger and the reports. Row 5 is the safety net for unpaid days; row 6 handles theft and tampering. Bold grey boxes are decisions.

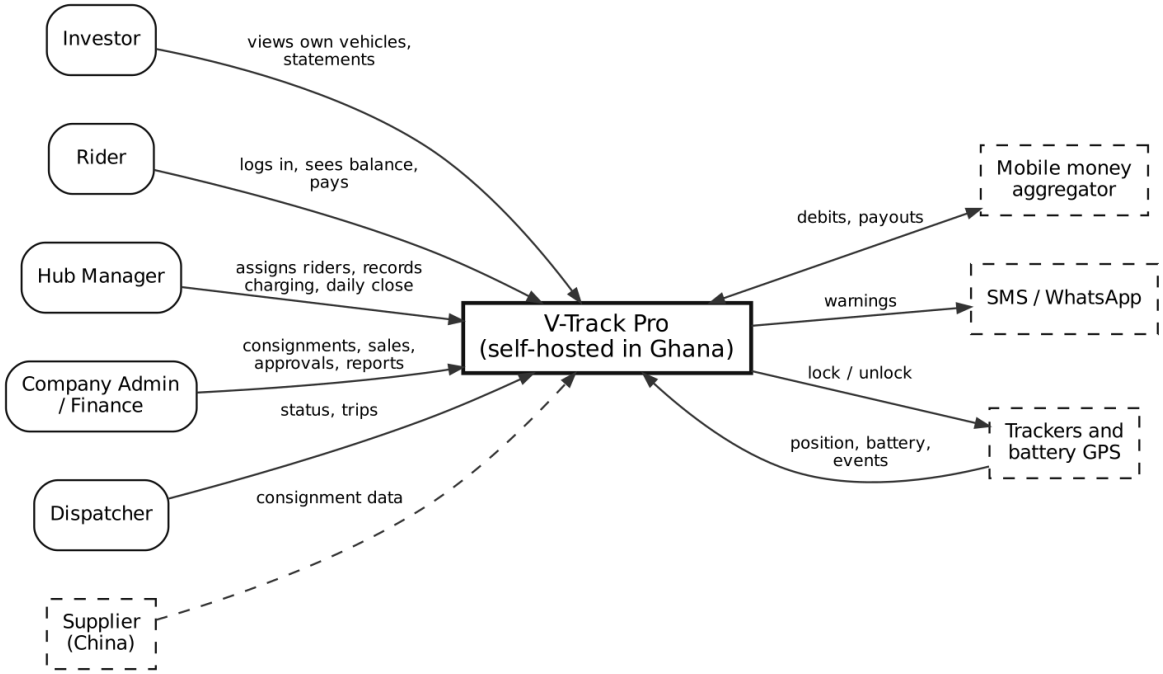
1. Purpose and Design Decisions

This document is the blueprint for the whole system. It defines who uses it, what it stores, how each thing moves through its life, how money flows, and which rules the software enforces. Building follows this model; if the model is wrong, the software will be wrong.

The ten design decisions that shape everything

#	Decision	Why
D1	The company is one tenant. Investors, riders and hub managers are users and records inside it, not separate tenants.	Matches the client's single business; keeps reporting simple.
D2	The battery is its own asset, separate from the vehicle.	Batteries are removable and swappable (72V 100Ah); they move between vehicles and are charged and stolen on their own.
D3	All money lives in an append-only ledger.	Every figure on every report can be traced to entries that were never edited.
D4	Money is stored as whole pesewas (integers) in GHS.	Avoids rounding errors that creep in with decimals.
D5	One contract model covers outright, lease and Work-and-Pay, with a schedule for each.	One place for price, balance and arrears, regardless of mode.
D6	A daily settlement is the unit of accounting per rider, vehicle and day.	Matches how the client thinks: daily collection, weekly and monthly roll-ups.
D7	Locking follows the safety rules already built: grace period, stopped vehicle, audit.	Protects riders and the company from wrongful or dangerous locks.
D8	Hub-level data isolation: a hub manager sees only their hub.	Limits damage from mistakes or misuse; supports the security goal.
D9	Every external payment has a unique reference and is processed once.	Mobile money callbacks can arrive twice or late.
D10	Reuse the existing platform: vehicles, devices, riders, payments, geofences, alerts, commands, audit.	Less to build before the end-of-November MVP.

2. System Context and Actors



Solid boxes are people; dashed boxes are outside systems. Everything in the middle is hosted in Ghana.

Actors

Actor	Responsibility in the system
Company Admin	Owns the business view: consignments, pricing, contracts, approvals, users, all reports
Finance Officer	Reconciles payments, reviews settlements and arrears, handles payouts to investors
Hub Manager	Runs one hub: assigns riders, records charging and swaps, closes the day, handles exceptions
Dispatcher	Watches vehicles and trips; checks status; later assigns trips by WhatsApp
Investor (Owner)	Reads own vehicles, balances, payback and statements; no access to others
Rider (Operator)	Sees own assignment, daily target and balance; pays and receives notices
Auditor	Read-only access to records and the audit trail

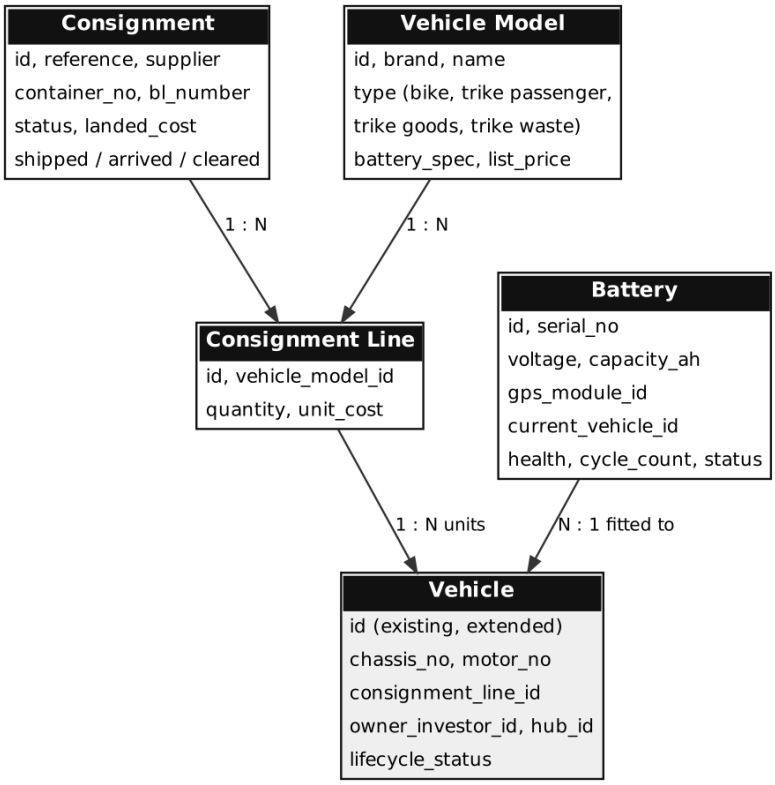
A rider who buys a vehicle through Work-and-Pay is both an Owner and a Rider. The model allows one person to hold both roles.

3. Modules and What Is Built

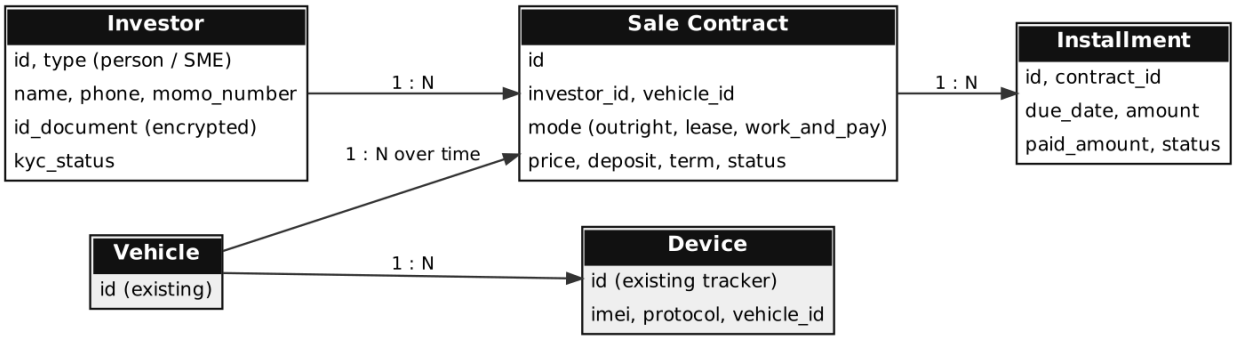
Module	Entities	Existing	Phase
A. Tracking and security	Vehicle, Device, Battery, Geofence, Alert, Command	Mostly built; Battery is new	1
B. Sales and consignments	Supplier, Consignment, Line, Vehicle Model, Owner, Sale Contract, Installment	New	1
C. Hubs and operations	Hub, Charge Point, Rider, Assignment, Swap, Charge Event	Rider exists; rest new	1 (basic), 2 (automation)
D. Finance	Payment, Daily Settlement, Ledger Account, Ledger Entry	Payment log exists	1 (core), 2 (wallet, aggregator)
E. Reporting	Daily, weekly, monthly, investor, consignment reports	Report framework exists	1 (core), 2 (more)
F. Waste collection	Estate, Waste Request, Pickup	New; needs design	3

4. Domain Model: Assets and Sales

Supply side: from supplier to a registered vehicle with a battery and tracker.



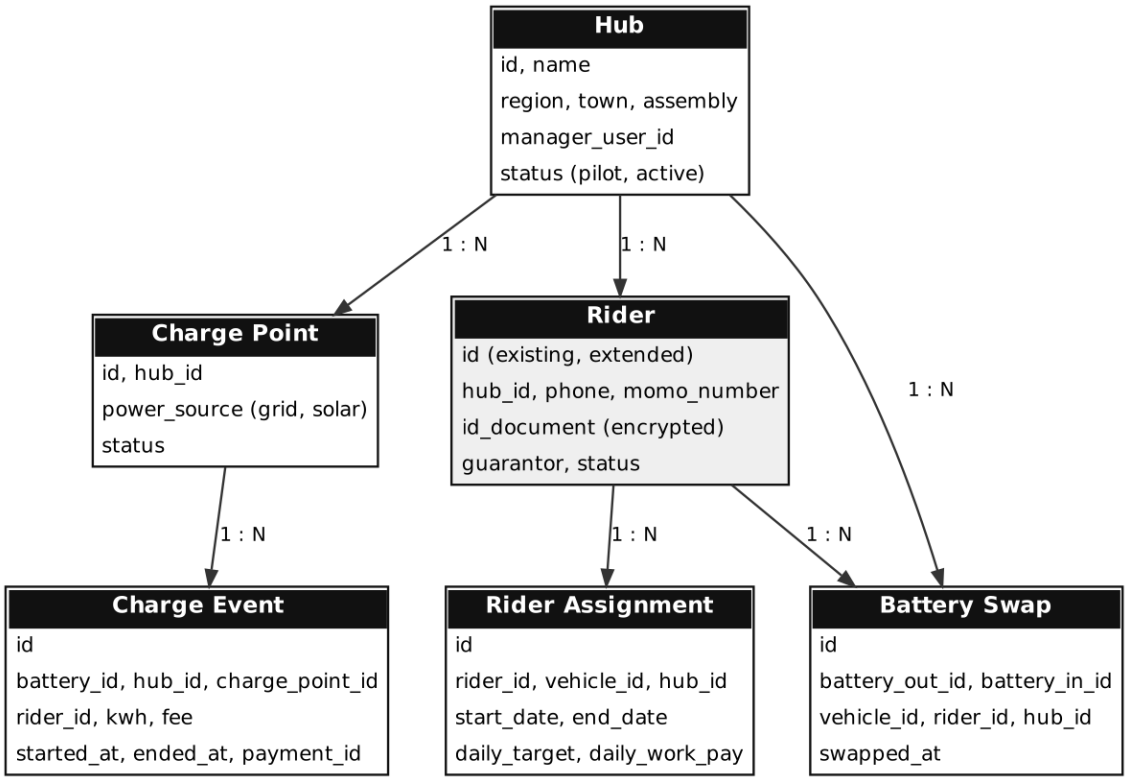
Sales side: who owns a vehicle and how it is paid for.



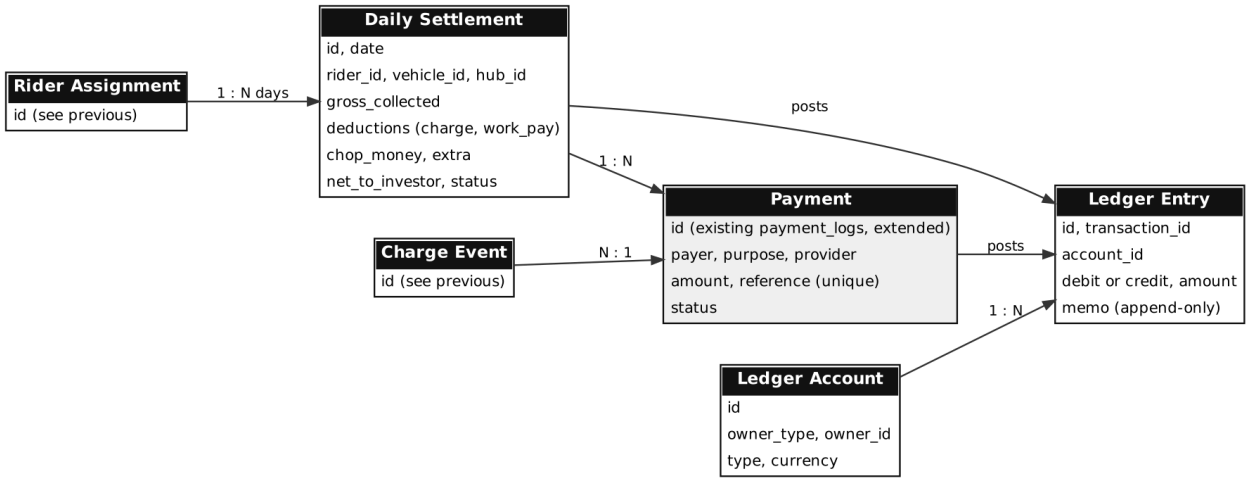
Grey boxes already exist in the platform and are extended. White boxes are new.

5. Domain Model: Operations and Money

Operations: hubs, riders, assignments, swaps and charging.



Money: daily settlement, payments and the ledger.



Grey boxes already exist and are extended. White boxes are new.

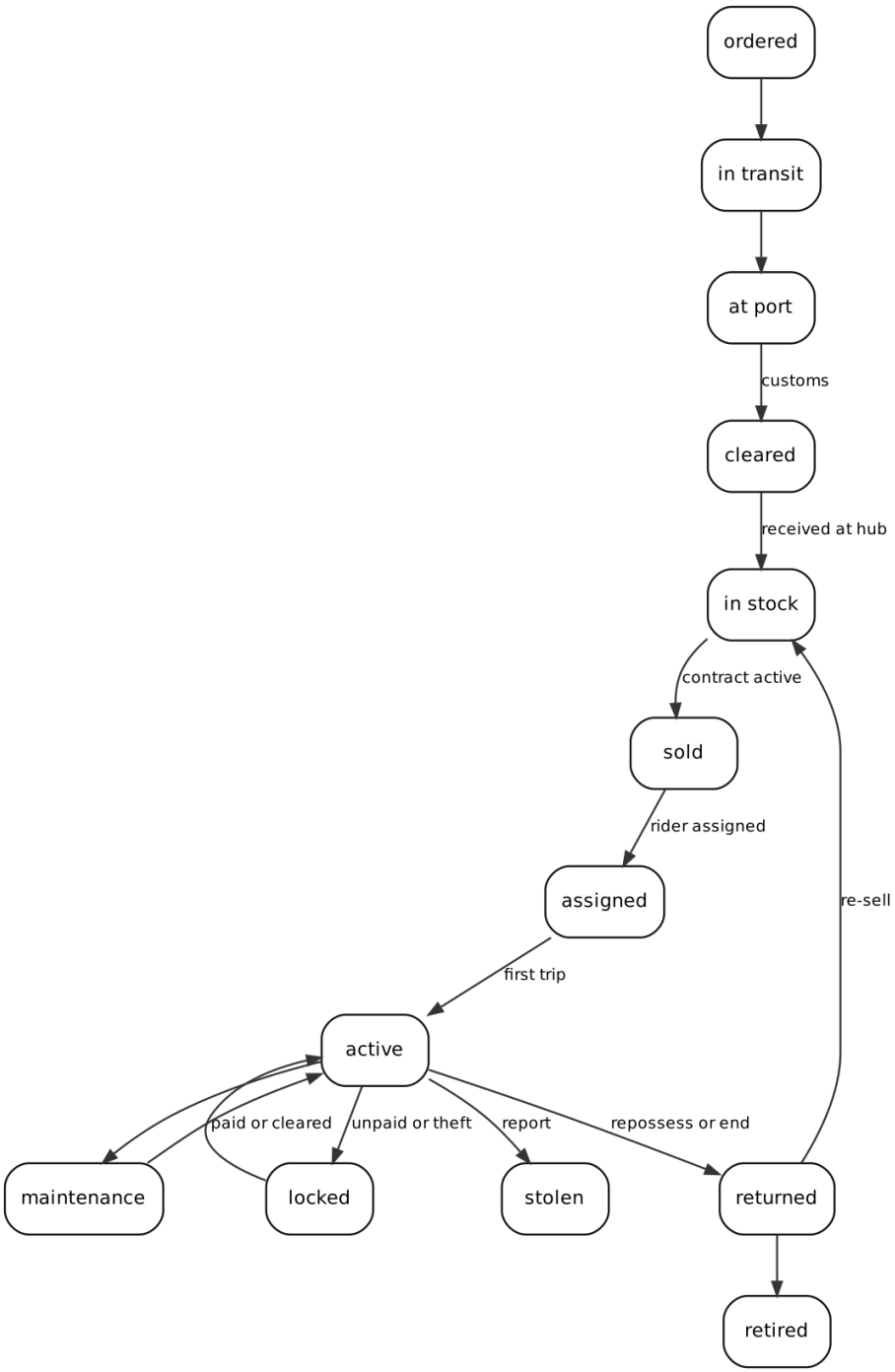
6. Entity Dictionary

Entity	Purpose and key rules	Phase
Supplier	Overseas maker. Name, country, contact. Informational.	1
Vehicle Model	Catalogue entry: brand (Huodi, YJJ, Bajaj, Apsonic), type, battery spec, list price.	1
Consignment	One container. Tracks status from order to closed, with landed cost (goods, freight, duty, clearing).	1
Consignment Line	Model and quantity within a container, with unit cost. Creates the individual vehicle records on receipt.	1
Vehicle (extended)	One physical unit with chassis and motor numbers, its consignment line, owner, hub and lifecycle status.	1
Battery	Serial, voltage, capacity, GPS module id, health, cycle count, and the vehicle it is fitted to now.	1
Device (existing)	The tracker, linked to a vehicle. A battery GPS module may be a second device source.	1
Owner / Investor	Person or SME. Contact, MoMo number, ID document (encrypted), KYC status.	1
Sale Contract	Links owner and vehicle. Mode, price, deposit, term, status. Drives the installment schedule.	1
Installment	One due payment with due date, amount, paid amount and status. Lease: fixed. Work-and-Pay: generated daily.	1
Hub	A base in a region, town or assembly with a manager and status (pilot, active, closed).	1
Charge Point	A charger at a hub. Power source (grid or solar) and status.	2
Rider (extended)	Operator with hub, phone, MoMo number, ID document (encrypted), guarantor and status.	1
Rider Assignment	Rider to vehicle at a hub for a period, with daily target and daily Work-and-Pay amount.	1
Battery Swap	Records battery out and in, vehicle, rider, hub and time.	1 (manual), 2 (automatic)
Charge Event	A charge session: battery, hub, kWh, fee, rider and the payment that settled it.	1 (manual entry)
Daily Settlement	Per rider, vehicle and day: collected, deductions, chop money, extra, net to owner, status.	1
Payment (extended)	Existing payment log, extended with purpose (charge, installment, deposit), payer and unique reference.	1
Ledger Account	A holder of value: rider, owner, company, hub cash, aggregator clearing.	1
Ledger Entry	Debit or credit on an account within a transaction. Append-only.	1

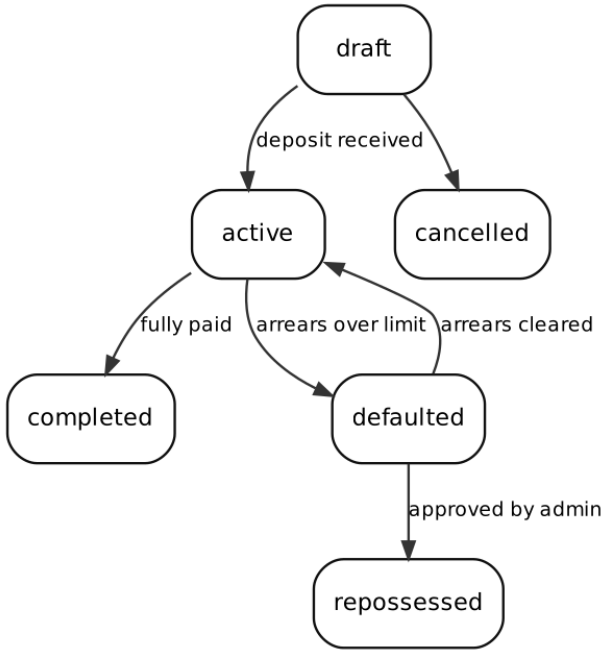
7. Lifecycles

Each entity moves through fixed states. The software allows only the arrows shown, so no record can jump to a nonsensical state.

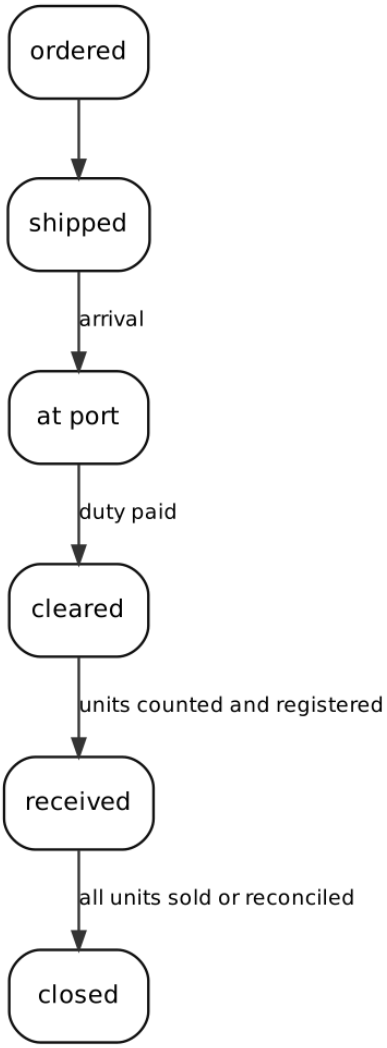
7.1 Vehicle



7.2 and 7.3 Contract and consignment



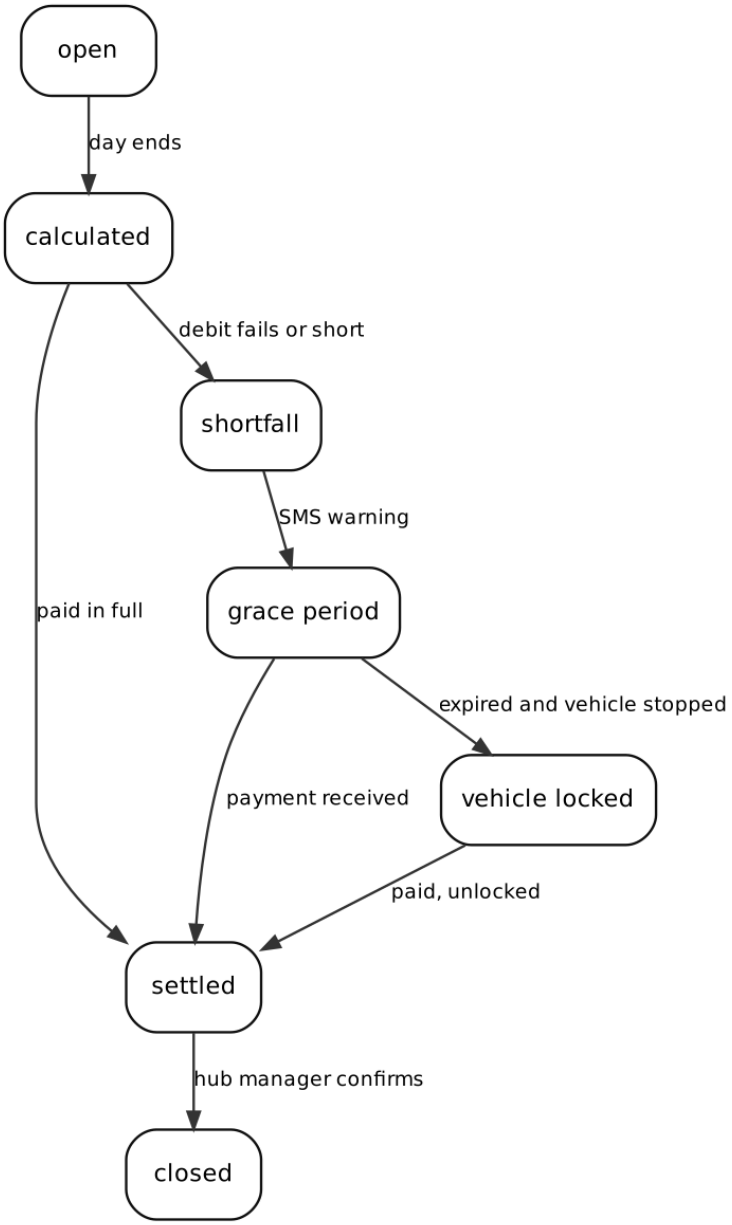
7.2 Sale contract



7.3 Consignment

- **Contract defaults** are a decision for a person: the system flags arrears and proposes default; an admin approves repossession. It never repossesses automatically.
- **A consignment closes** only when every unit is sold, in stock, or written off with a stated reason.

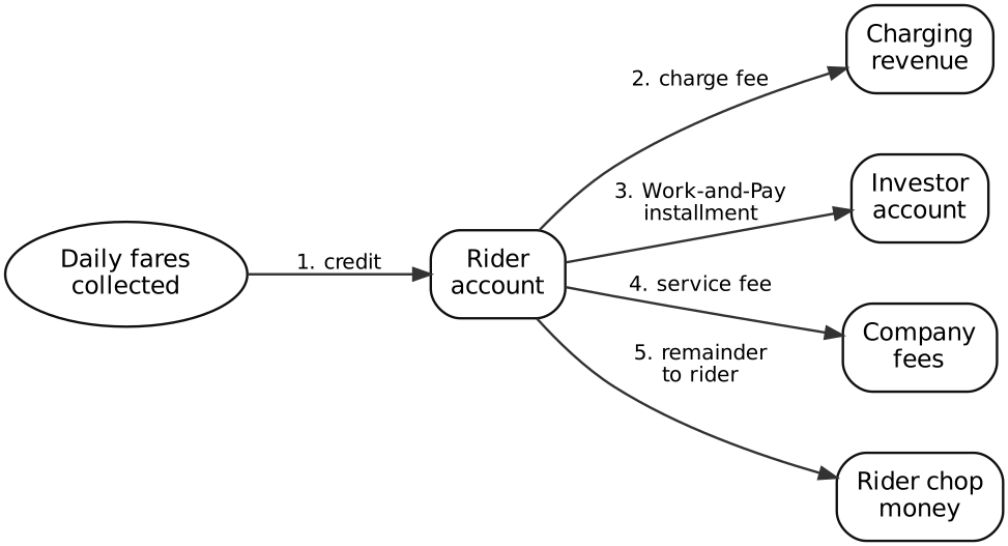
7.4 Daily settlement and the safe lock



- This reuses the Work-and-Pay logic already built: warning, grace period (default 12 hours), speed check, delayed retry, lock.
- A vehicle that is moving above 5 km/h is never locked; the system waits and retries.
- Payment at any point before the lock ends the shortfall. After a lock, payment triggers the unlock.

8. Money Model

Every movement of money becomes ledger entries: one or more debits and credits that sum to zero. Reports are built from the ledger; nobody edits a past entry. A mistake is fixed by a reversing entry.



Order of deduction from a rider's daily collection (assumption A2, to be confirmed)

Accounts

Account	Holds	Type
Rider account	Fares collected and amounts owed or released	Per rider
Owner account	Share due to the owner; installments received	Per owner
Company fees	Service fees retained by the company	Company
Charging revenue	Battery charging fees	Company
Hub cash	Cash held by a Hub Manager awaiting deposit	Per hub
Aggregator clearing	Mobile money in transit between aggregator and bank	Company

Worked example (illustrative figures, not client data)

Step	Amount (GHS)	Entry
Rider collects fares for the day	300.00	Credit rider account
Charging fee for the battery	25.00	Debit rider, credit charging revenue
Daily Work-and-Pay remittance	80.00	Debit rider, credit owner or contract
Company service fee	10.00	Debit rider, credit company fees
Remainder is the rider's chop money	185.00	Debit rider, credit rider chop

Check: 25 + 80 + 10 + 185 = 300. Settlement is complete when every cedi collected is assigned to an account.

Formulas used in reports

- **Outstanding balance** = price - deposit - sum of paid installments.
- **Weekly net to owner** = sum of the owner's share over the seven days.
- **Payback weeks** = price / average weekly net to owner.
- **Arrears** = sum of unpaid due amounts; **arrears days** = number of days with a shortfall.
- **Hub collection rate** = amount collected / amount due, over the period.

9. Business Rules

Rule	Statement
BR1	A vehicle has at most one active contract and one active rider assignment at a time.
BR2	A rider has at most one active assignment.
BR3	A battery is fitted to at most one vehicle; every change is recorded as a swap.
BR4	Money is stored in whole pesewas. No decimals in storage or calculation.
BR5	Ledger entries are never edited or deleted. Corrections are reversing entries.
BR6	Every payment has a unique reference. A repeat callback is ignored.
BR7	A daily settlement is calculated once per rider, vehicle and day. A recalculation creates a new version and keeps the old one.
BR8	A lock requires: grace period passed, vehicle stopped (5 km/h or less), audit entry. A manual lock needs approval.
BR9	Unlock is automatic on full payment, or manual by a Hub Manager with a reason.
BR10	A contract is flagged for default after N days of arrears (configurable). Repossession needs admin approval.
BR11	A Hub Manager sees and edits only records of their own hub.
BR12	An Investor sees only own vehicles, contracts and statements.
BR13	ID documents and MoMo numbers are encrypted and visible only to roles that need them.
BR14	A consignment closes only when all units are sold, in stock, or written off with a reason.
BR15	A sale priced below landed cost needs admin approval.
BR16	The grace period is configurable per contract or hub; default 12 hours.
BR17	A rider exception (sickness, breakdown, leave) pauses the daily target and the lock for a stated period, approved by the Hub Manager and logged. This rule is proposed and awaits the client's decision.

10. Roles and Permissions

F = full, R = read, H = own hub only, O = own records only, A = approve, dash = no access.

Area	Admin	Finance	Hub Mgr	Dispatch	Investor	Rider	Auditor
Consignments and cost	F	R	-	-	-	-	R
Sale contracts	F	F	R (H)	-	O	O	R
Vehicles and live map	F	R	F (H)	R	O	O	R
Riders and assignments	F	R	F (H)	R	R (own vehicles)	O	R
Charging and swaps	F	R	F (H)	-	-	O	R
Daily settlement	F	F	F (H)	-	O	O	R
Ledger and payments	F	F	R (H)	-	O	O	R
Lock and unlock	A	-	Request	-	-	-	R
Reports	F	F	H	R	O	O	R
Users and roles	F	-	-	-	-	-	R

Row-level rules: a Hub Manager is tied to one hub; an Investor to the vehicles they own; a Rider to their own assignment. Every read of personal data and every lock command is written to the audit log.

11. Reporting Model

Report	Audience	Period	Contents
Hub daily close	Hub Manager, Admin	Daily	Riders active, collected, due, shortfalls, charging fees, batteries out and in
Rider statement	Rider, Hub Manager	Daily, weekly	Collections, deductions, chop money, balance, route summary
Investor statement	Investor, Admin	Weekly, monthly	Vehicle activity, owner share, installments, outstanding balance, payback progress
Arrears and defaults	Finance, Admin	Daily	Riders and contracts in shortfall, days and amount, lock status
Consignment profitability	Admin	On demand	Landed cost, units sold, revenue, margin, units in stock
Fleet and battery health	Admin, Technician	Weekly	Utilisation, idle vehicles, battery health and cycles, charge events
Weekly and monthly summary	Admin, client	Weekly, monthly	Totals per hub, region and vehicle type; trends

12. The Daily Cycle, Step by Step

Time	Step	Records created
Morning	Hub Manager confirms active riders and vehicles; riders start (ID session); a charged battery is issued	Rider session, battery swap
During the day	Vehicles report position and battery; trips and routes are built; alerts raised for theft or tamper	Positions, trips, alerts
Charging	Battery is returned and charged at the hub; fee noted	Charge event, payment or charge debt
6 PM	System requests the daily debit through the aggregator	Payment (pending)
Result	Success: settled. Failure: SMS warning, grace period	Ledger entries, settlement state
Night	Settlement calculated; hub close prepared; unpaid items flagged	Daily settlement, reports
Next day	Hub Manager confirms close; locked vehicles released on payment	Closed settlement, unlock audit

13. MVP Build Order for End of November

About three weeks remain. The order below delivers a working pilot hub first and leaves automation for later.

Week	Build	Reuses
1	Owners, hubs, vehicle models, consignments and lines, vehicle registration with lifecycle	Vehicles, devices, auth, roles, audit
2	Sale contracts and installments; rider assignments; batteries and manual swap and charge entry	Riders, rider sessions
3	Daily settlement, ledger, weekly and monthly reports; connect Work-and-Pay to settlement; hub-level access	Work-and-Pay processor, payment logs, command safety
After	Real aggregator, battery GPS integration, wallet, WhatsApp, NFC, waste collection	-

Honest note: this is a tight plan. If time runs out, cut in this order: monthly reports, charge events, battery swaps. Keep consignments, contracts, assignments and daily settlement.

14. Assumptions to Confirm with the Client

#	Assumption	Question for the client
A1	An owner may be an investor who hires a rider, or the rider who buys through Work-and-Pay.	Is that right, and does an investor share each day's collection with the rider?
A2	Deductions from daily collection follow the order: charging, remittance, company fee, rider remainder.	What is the correct order and what are the amounts per vehicle type?
A3	The company collects fares through mobile money wallets, not cash.	Is some cash collected at hubs by the Hub Manager?
A4	The grace period is 12 hours.	Is that acceptable, and is there a rider exception policy (BR17)?
A5	A lease has a fixed schedule; Work-and-Pay has a daily target.	What are the deposit, term and amounts for each mode?
A6	Money is in Ghana Cedis only.	Any other currency needed, for example for supplier costs?
A7	The battery's own GPS lock may stay separate from our lock.	Can the supplier provide the module's protocol, or will we fit our own tracker?